

COMP 3300 – Fall 2025
Assignment 3, 10 pts.
Due: Sunday, Sept 7th @11:55pm

Objectives

- Recreate an existing console application into a Windows Forms Application

Notes

- Please read through this entire document before you begin, so you know all the requirements for the program at the outset.
- **Any program that does not compile will receive a zero (0). Partial credit is not possible for any program that does not compile.**

Background

This exercise is one that often happens in development – a pre-existing application exists in one framework/type of application and the functionality must be duplicated to a new form/application. For this assignment we are going to focus on migration of Assignment 1 code from a ***Console Application*** to a ***Windows Forms Application*** (WinForms).

You have complete autonomy to design the layout/look of your forms as you see fit, provided the requirements of the program are met. This assignment we are simply focused on getting familiar with working within a GUI framework and not designing perfect user interfaces. You are not required to add in any menus/etc.

Getting Started

1. Start Visual Studio 2022 and create a new C# Windows Forms Application that is .NET 8 , i.e., it can run on Windows (WinForms will NOT work on Linux/Mac OS). This should **NOT** be a .NET Framework application.
 - a. Name the project *FirstnameLastnameA3*, e.g., SallieMaeA3.
2. Build and run the application to verify it runs. It should display a Window named “Form1” that is empty (no controls).
 - a. You will notice that Program.cs already has a Main method defined. **Do NOT modify this code.** Most, if not all, of the code changes will be within the Form1 and its code behind.

Requirements

1. Set the form title (title bar) to be *Your Full Name* Assignment 3
2. Since there is not a “Console” window available, all output back to the user must be displayed within an appropriate control. If a requirement below communicates back to the user of the program in some way, it should be displayed in that appropriate control.
3. Design the Form to contain the following:
 - a. Users enter two words.
 - There MUST be two words entered and
neither can be null/empty/whitespace
and users should be notified of any
issues with the words they entered.

The following requirements will also necessitate the use of appropriate control(s) to “trigger” their execution. The application will perform that particular action when that control triggers it. You will need to locate the appropriate control and set it up to execute that behavior. In addition, you should consider how the control appears to the user (*hint*: titles). You may also have to modify output the message to the user based on the action being performed (for example the palindrome behavior in assignment 1 was added to console output describing both words).

[Microsoft provides a rich API about controls.](#)

- b. Program will provide metrics on the two words entered (Word fun 1 in Assignment 1).
- c. Program will print the “word square” (Word Fun 2 in Assignment 1).
- d. Program will display a word pyramid (Word Fun 3 in Assignment 1)
- e. Program will output if either word is a palindrome (Word Fun 4 in assignment 1)
- f. Program will notify users of the letters that appear in both words. This is new behavior and not part of any previous assignment. For example, if the

words “One” and “Orange” are entered, the user would be notified that O, N, E are present in both.

- g. Program will allow users to enter numbers spelled out (Like “One” and “Four” – case insensitive), and then allow those to be converted to numeric values and multiplied. The resulting value should be displayed to the user.
- You MUST use a method to “parse” to integers (you do not need to account for double/decimal).

For example – if a user enters “four” and “nine” the value displayed will be 36.

Hint: [Int has a few methods available to “Parse”](#) one is much better suited to this task. While any can work I would suggest reading closely about how each does what it does – if a user enters a non-number as a word and it can’t be parsed there can be side effects.

Since WinForms continue until the program is closed (like a standard windows application) there is no need to duplicate the looping feature of the Assignment 1

Grading breakdown

- 10 pts. – Exercise completed successfully with at most only a minor error and follows best programming practices.
- 7 pts. – Exercise is missing some functionality or has some errors in it and mostly follows best programming practices.
- 4 pts. – Exercise is missing most of the major functionality or has major errors or best programming practices not followed.
- 0 pts. – Exercise not submitted, does not compile, or barely started.

Submission

Make sure you clean your solution (Build → Clean Solution) and then zip up your project which needs to include the solution file into a ZIP file called *YourNameA3.zip* and submit the assignment by the due date.